Engineering Epistemic Conditions

Designing conditions for inquiry without pretending to manufacture truth

SymK

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Contents

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Purpose and audience

Purpose. This paper explains the object, practice and limits of Knowledge Engineering in SymK. It offers a self-contained newcomer path, followed by the historical, conceptual and engineering detail needed for deeper study.

Audience. People who build or govern knowledge-intensive systems, including domain experts, designers, engineers, reviewers and affected participants. The opening assumes no prior SymK history or formal knowledge-representation training.

The question this paper answers. If Knowledge cannot be manufactured like an ordinary product, what exactly does Knowledge Engineering design and govern?

Authority and limits. This is a governed explanatory paper, not a constitutional norm, final semantic package, formal specification or implementation contract. Exact Stage-Accepted decisions control if wording here and those decisions differ.

SymK in plain language

SymK is a general framework and engineering discipline for responsible Knowledge Engineering. It helps people and other capable systems work together on difficult questions while keeping claims, evidence, authority, responsibility, use, consequences and correction visible.

In ordinary language, Knowledge Engineering is the disciplined work of making Domain knowledge expressible, organized, connected to evidence, reviewable, maintainable, challengeable and usable by people and computational systems. Its artifacts may include conceptual models, vocabularies, ontologies, knowledge bases, reasoning structures, provenance records, validation rules and workflows. It may draw on knowledge management, data engineering, software engineering and AI engineering, but it is not identical to any one of them.

Within SymK, Knowledge Engineering is the integrative center of the paper set and epistemic conditions are its direct engineering object. Knowledge, Intelligence and Formation are constitutive dimensions it must preserve; Education and Training are conditional formative practices rather than universal requirements. This describes functional architecture, not constitutional rank. The architecture remains open to future supporting concepts admitted through evidence-controlled governance when a necessary distinction, relationship or failure mode cannot otherwise be preserved.

SymK began with Human-AI cooperation and still treats Domain Intelligence Amplifiers as a principal realization. Its scope is now broader: it can examine governed cooperation among different forms and Bearers of Intelligence. This broader identity does not make every activity part of SymK, erase Human standing or give a technical system authority merely because it is capable.

Why this paper matters. A well-structured database, ontology, model or workflow may still contain false claims, hide uncertainty or authorize unsafe reliance. Knowledge Engineering must therefore design the conditions for inquiry, evidence, representation, assessment, challenge, authorized use, correction and learning without pretending to manufacture truth.

Two reading paths

Short path - about fifteen minutes. Read this opening, the Abstract, section 1 and the Conclusion. That path states the problem, the central answer, the practical consequence and the limits without requiring the full evolution history.

Full path - for deeper knowledge. Continue through every numbered section, including the counterexamples, distinctions, engineering implications, dissent and references. Those sections support scrutiny and specialist use; they are not prerequisites for understanding the main claim.

The optional *SymK Reader Orientation and Shared Glossary* introduces recurring terms across all four Foundation Papers. This paper nevertheless defines the terms required for its own main argument.

One example to carry through the paper

For the bridge case, Knowledge Engineering concerns the whole epistemic arrangement: sensors and calibration, inspection roles, claim and evidence records, model provenance, representation choices, review procedures, competent operational authority, challenge routes, reliance decisions, consequence monitoring and correction. The database and AI output are parts of that arrangement. Neither is a container whose presence proves that Knowledge exists.

The example is illustrative. It does not create a universal workflow, legal conclusion or operational rule.

Essential terms for this paper

- **Knowledge Engineering:** the design, construction, governance, evaluation and evolution of conditions that affect responsible epistemic work.
- **Epistemic conditions:** participants, processes, roles, resources, representations, controls, environments and feedback that enable or impair inquiry.
- **Representation:** an expression made for a purpose; it must not be confused with reality, the represented subject or Knowledge itself.
- **Projection:** a selective view or reduction of a richer source, with purpose and loss made visible.
- **Three Views:** Human, Scientific and Engineering perspectives that must remain connected without one silently replacing the others.
- **Responsibility and Authority:** separate, scoped relations governing who must answer or act and who is competent to decide.

Capitalization marks terms that have a controlled SymK use. These short explanations support reading; the Stage-Accepted decisions remain the controlling conceptual source.

Abstract

SymK is a general framework and engineering discipline for responsible Knowledge Engineering through governed cooperation among different forms and Bearers of Intelligence. Its object is not truth or Knowledge as a manufactured product. It is the set of **epistemic conditions** that shape how Knowledge is pursued, achieved, attributed, expressed, preserved, communicated, assessed, challenged, applied and revised within declared Contexts and Scopes.

This limit matters. A claim, document, model, graph, database, generated output, assessment, approval or record does not become Knowledge through storage, structure, inference, Authority, adoption or naming. Knowledge Engineering can enable, constrain, preserve, expose or impair epistemic achievement. It cannot declare truth into existence or guarantee that any Bearer knows.

SymK therefore treats Knowledge Engineering as the design, construction, governance, evaluation and evolution of arrangements involving participants, processes, roles, resources, representations, controls, environments and feedback. Every representation is selective. Formal correctness is local to declared premises and semantics. Generative systems produce candidates rather than warrant. Evidence may challenge the claims, premises, applicability, consequences or decisions made under authority, while change to an authority or governed artifact follows the competent procedure. Urgent evidence calls for prompt consideration by competent operational or external authority; it does not invent an emergency power or duty.

The detailed paper presents a branching and recursive lifecycle, Representation and Projection boundaries, Human/Scientific/Engineering Views, typed Authority and Responsibility, the lowest-adequate-Bearer rule and a

non-container review envelope. These guide later design without defining one universal schema, workflow, ontology, metric, certification or implementation package.

Review status and source control

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Authority boundary: Governed explanatory source; not a constitutional norm, final vocabulary or contract package, formal specification, authority allocation or implementation

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1. Why Knowledge Engineering is Foundational to SymK

SymK has layered identities that must remain visible together:

- historically, a protocol for symbiotic cooperation between Human Intelligence and Artificial Intelligence;
- principally, though not exhaustively, an orientation toward Domain Intelligence Amplifiers; and
- more broadly, an engineering discipline and general framework for responsible Knowledge Engineering through governed Cooperation among heterogeneous forms and Bearers of Intelligence.

None of these layers claims all Knowledge Engineering, all Human–AI relations, all systems, or all cooperation. A governed knowledge undertaking is useful as a non-reifying analytical center: it lets SymK ask how purposes, Questions, participants, claims, Representations, decisions, actions, consequences, and revisions relate without inventing a new Foundational Entity or automatic collective Bearer.

The integrated baseline contains seventeen Foundational concepts:

1. Identity;
2. Context;
3. Scope;
4. Domain;
5. Representation;
6. Knowledge;
7. Intelligence;
8. Knowledge Engineering;
9. Cooperation;
10. Formation;
11. Responsibility;
12. Amplification;
13. Domain Intelligence Amplifier;
14. Question;
15. Reasoning;
16. Answer; and
17. Value.

Knowledge Engineering relates these concepts through deliberate construction. It asks how participants with different substrates, capabilities, roles, vocabularies, Representations, temporal positions, access to Evidence, Authorities, Responsibilities, and affected interests can pursue epistemic work without collapsing their differences.

Without a disciplined boundary, recurring reductions follow:

- ontology construction becomes the whole of Knowledge Engineering;
- stored assertions become “facts” or Knowledge by label;
- expertise becomes extractable content;
- formal validity becomes truth, adequacy, or permission to rely;
- a model output becomes an epistemic achievement;
- one validator substitutes for every other assessment dimension;
- Knowledge management becomes document movement;
- governance is appended after a design has already shaped access, visibility, classification, and control;
- practical, tacit, embodied, disputed, or distributed contributions disappear because they resist one formalism; and
- consequences for participants and affected subjects remain outside review.

Knowledge Engineering is therefore Foundational because SymK needs a stable concept for engineering epistemic conditions. A Foundation Paper explains that concept; it does not Ratify constitutional language, allocate external Authority, or certify an implementation.

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2. Exact accepted definition and limiting principle

> **SymK shall govern Knowledge Engineering as a foundational concept. Knowledge Engineering is the disciplined design, construction, governance, evaluation, and evolution of epistemic conditions: arrangements of participants, processes, roles, resources, representations, controls, environments, and feedback that materially affect how Knowledge is pursued, achieved, attributed, expressed, preserved, communicated, assessed, challenged, applied, and revised within declared Contexts and Scopes.**

An **Epistemic Condition** is a supporting distinction. It identifies something that can materially affect epistemic work. It is not automatically:

- an Entity or container;
- a new Foundational concept;
- a Knowledge Bearer;
- Knowledge, a claim, a Ground, or Evidence;
- an epistemically good or responsible condition;
- sufficient or necessary for success; or
- a guarantee of truth, Knowledge, justified reliance, beneficial action, or Value.

Knowledge Engineering can enable, constrain, preserve, expose, or impair epistemic achievement. A well-designed process can still fail. A poor process can sometimes be followed by a true claim. Neither outcome erases the need to assess the condition, claim, Grounds, Bearer, Context, Scope, Authority, action, and consequence separately.

The limiting principle is strict:

> **Knowledge Engineering does not manufacture truth or create actual bearing by declaration. No claim, Document, model, ontology, graph, database, semantic or software package, generated output, assessment, approval, or operational record becomes Knowledge merely through storage, structure, inference, Authority, adoption, naming, or inclusion in SymK.**

“Responsible” characterizes intended and assessed conduct under applicable criteria; it is not an automatic consequence of using a governed process. “Shareable,” “actionable,” or “self-correcting” similarly require separate Evidence and cannot be built into the existence condition of Knowledge Engineering.

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3. The inherited discipline and its bounded lessons

3.1 Expert systems and task-specific structure

Early expert systems such as DENDRAL and MYCIN showed that specialized performance depended heavily on domain-specific structures, not only on a universal reasoning procedure. Feigenbaum used *Knowledge Engineering* for applying AI principles and tools to difficult problems requiring expert Knowledge, with acquisition, Representation, use, and explanation as central concerns [Feigenbaum 1977; Buchanan and Shortliffe 1984].

The Knowledge engineer elicited rules, exceptions, categories, heuristics, and strategies; encoded them; observed failures; and revised the artifact. This remains part of the field’s history, not a complete contemporary definition.

3.2 The acquisition bottleneck

Experts omit what has become routine, disagree, reconstruct reasons retrospectively, and rely on perceptual or practical discrimination that may not be fully articulable. Interviews produce situated testimony about practice, not transparent copies of an internal mechanism. Anthropological study of Knowledge Engineering also showed that elicitation is socially organized and shaped by roles, expectations, and institutional settings [Forsythe 1993].

The bottleneck is therefore not only expensive elicitation. It exposes a defective transfer metaphor: expertise is not a container whose contents can be losslessly extracted.

3.3 From transfer to modeling

KADS, CommonKADS, MIKE, and related methods shifted the field toward constructed models of organizations, tasks, agents, communication, inference, and Domain content [Wielinga, Schreiber, and Breuker 1992; Studer, Benjamins, and Fensel 1998; Angele et al. 1998; Schreiber et al. 2000]. A knowledge base became an engineered model for purposes and tasks rather than a reproduction of expert cognition.

This shift made assumptions, abstractions, interfaces, roles, and iteration more visible. It remains a methodological contribution, not proof that a model is true or that its user knows.

3.4 Ontologies, formal semantics, and Knowledge graphs

Computational ontology engineering made conceptualizations explicit and reusable [Gruber 1993]. OWL 2 supplies declared formal semantics and reasoning conditions for ontology artifacts [W3C 2012]. Knowledge graphs support linking, identity management, traversal, integration, search, and inference across structured assertions [Hogan et al. 2021].

These are substantial capabilities. Still, agreement on vocabulary does not make every category appropriate, every premise true, or every consequence just. A Knowledge graph may carry supported, disputed, superseded, generated, hypothetical, or erroneous claims. Its name does not decide their epistemic status.

3.5 Machine learning and generative systems

Machine learning and generative systems expand the materials of Knowledge Engineering: datasets, labels, learned parameters, prompts, retrieval contexts, tools, generated candidates, evaluations, review arrangements, and operational feedback. They relocate parts of the acquisition bottleneck toward source selection, measurement, model Formation, grounding, adjudication, monitoring, and maintenance.

> **Generative systems can accelerate proposition, transformation, comparison, and inquiry. They do not automate epistemic warrant, Knowledge attribution, Authority, Responsibility, or permission to rely.**

This historical section is explanatory. Its sources inform the account but do not settle SymK semantics.

4. What Knowledge Engineering can engineer

Knowledge Engineering acts on epistemic conditions and transition arrangements. Depending on Context and Scope, it can engineer:

- purpose and Question-framing procedures;
- participant, contributor, assessor, challenger, decision, and affected-subject interfaces;
- elicitation, observation, inquiry, measurement, comparison, and acquisition arrangements;
- claim, Ground, Evidence, Provenance, uncertainty, disagreement, and status records;
- vocabularies, Documents, models, rules, ontologies, graphs, examples, simulations, datasets, and interfaces;
- reasoning, retrieval, computation, generation, explanation, and evaluation processes;
- assessment, review, challenge, correction, escalation, and Reliance Authorization procedures;
- access, security, privacy, versioning, supersession, retirement, and accountability controls;
- memory and capability-continuity arrangements;
- operational observation and consequence-return routes; and
- conditions that may contribute to Formation, without presuming Learning, Training, or Education occurred.

It cannot directly engineer:

- truth by declaration;
- Knowledge by storing a Representation;
- understanding by producing explanation-shaped text;
- competence by documenting a procedure;
- consensus by erasing disagreement;
- a collective Bearer by assembling contributors;
- responsible reliance by assigning a score or approval label;
- Education merely by exposing a participant to an interface; or
- beneficial Value merely by meeting a technical target.

4.1 The non-reifying object model

The relevant objects and events form a network, not a vertical substance stack:

```
Purpose / Context / Scope / Domain
        | frame and constrain
        +-----------------------+
        v                       v
Questions <--> Inquiry      Participants / roles / practices
     |             \          /       |
     |              \        /        | contribute, observe, challenge
     v               v      v         v
Candidate claims <--- Grounds / possible-Evidence / Provenance
     |  \                  |                |
     |   \ represent       | assessed as    | locates lineage
     |    v                v                |
     |  Representations / Projections <----+
     |       |       \
     |       |        \ reasoning, retrieval, computation, generation
     |       v         v
     +--> Assessments / Answers / recommendations
                  |             |
                  | competent   | may inform, never entail
```

```
              v              v
       Status / Reliance Authorization --> decisions / actions
              ^                               |
              |                               v
     challenge / review <--- observations / outcomes / consequences
              |                               |
              +--------- revision routes ---+
```

Reality and practice are not reduced to nodes in this diagram. Knowledge may be achieved by an identified Bearer in one of the accepted modes, but it is not generated by an arrow. Grounds can exist without being selected as Evidence. Evidence is a role relative to a specified object and criteria. Observed consequences are claims or observations that require interpretation and assessment before they can serve as Evidence. Revision may affect a claim, Representation, assessment, status, authorization, process, or artifact; competent procedure governs each change.

The model is recursive. A Question may arise from any branch. An Answer may become a claim in a later inquiry. A Representation may be reprojected. An assessment may be challenged. An action can alter the conditions from which later observations are made. No phase is universally first or last.

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5. Acquisition is construction, not extraction

5.1 Contributors are participants, not containers

A contributor’s statement is testimony or another possible Ground, not an unmediated export of Knowledge. Elicitation is shaped by Questions, examples, vocabulary, memory, incentives, role, Authority, Power, setting, and available Representations. The engineer also contributes assumptions through selection, omission, abstraction, translation, and conflict handling.

Acquisition is therefore model and claim construction under epistemic constraints. Contributor, source, author, Bearer, assessor, approver, decision-maker, affected subject, and responsible party remain distinct roles unless separately justified.

5.2 Tacit and practical Knowledge

People may know more than they can tell [Polanyi 1966]. Practical Knowledge may involve timing, perception, skilled movement, situated attention, participation, and responsiveness that are not exhausted by propositions. Explicit articulation can support teaching, coordination, critique, and computation, but it is a transformation. A procedure, example set, ontology, or model may contribute to capability without being identical to it.

5.3 Documents, datasets, and measurements

Documents preserve claims, observations, instructions, reasons, decisions, and records. They do not certify themselves. Review must preserve authorship, purpose, time, version, audience, jurisdiction, and relation to practice.

A dataset is produced through measurement, sampling, labeling, exclusion, transformation, and historical action. It is neither raw reality nor a neutral collection of facts. A learned regularity can support prediction or inquiry without thereby becoming Knowledge, causal explanation, or permission to act.

5.4 Generated candidates

Generated concepts, mappings, summaries, classifications, Questions, and proposed rules are candidates. Their source context, transformation, omissions, model and tool versions, uncertainty, conflict, possible Ground, Evidence role, assessment, and downstream use require governance. Human approval is not a warranty, and machine production is not disqualification; both are events whose epistemic significance must be assessed.

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6. Claims, Grounds, Evidence, Provenance, and status

Knowledge Engineering must keep at least the following distinctions visible:

- **Knowledge semantics:** what it means for a Bearer to know in an accepted mode;
- **Knowledge attribution:** a scoped, challengeable claim that an identified Bearer knows;
- **claim:** something asserted, denied, hypothesized, recommended, prescribed, or otherwise put forward;
- **Ground:** something that can bear a supporting, defeating, contextual, or explanatory relation;
- **Evidence:** a role attributed to a Ground relative to a specified object, question, criteria, assessor, Context, Scope, and time;
- **Provenance:** origin, custody, transformation, contribution, and lineage information;
- **assessment:** an evaluation under declared criteria by an identified assessor;
- **uncertainty and disagreement:** structured limits or conflict, not empty metadata;
- **status:** a governed label about an object under a procedure;
- **Reliance Authorization:** permission by competent Authority to rely for a declared purpose and Scope;
- **asset, record, Document, package, runtime object, and Projection:** engineered artifacts that do not become Knowledge through their existence.

Ground, Evidence, and Provenance do not independently guarantee truth, warrant, applicability, adequacy, or authorization. A source can be authentic but wrong. A method can be competent but misapplied. Evidence can support one object under one criterion and be irrelevant to another.

6.1 Evidence and competent procedure

Evidence may challenge claims, premises, Applicability, consequences, and decisions operating under any Authority. It does not automatically amend a governed artifact, alter or revoke an Authority or its source, or reverse a decision; such change follows competent procedure.

Urgent adverse Evidence triggers urgent consideration by competent operational or external Authority. This creates no emergency duty, containment or suspension power, decision right, remedy, or jurisdiction.

6.2 Standing and challenge

SymK preserves eleven typed Authority kinds under the accepted baseline. Affected-subject standing/challenge remains a distinct unresolved locus; it is not classified here as a settled twelfth Authority kind. Standing can make challenge and response procedurally relevant without automatically creating veto, legal standing, participation, remedy, or decision Authority. Domain and external authorities retain those determinations within jurisdiction.

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7. Representation, Projection, and the Three Views

Under the accepted structural kernel, **Representation** is Foundational and **Projection** is a supporting concept. A Representation is a situated form through which selected aspects of a subject are made available for interpretation, communication, Reasoning, storage, or action. Projection is the governed relationship through which selected dimensions of a source are expressed in another Representation, with transformations, additions, omissions, and known loss declared.

Representation is not preservation without remainder. A rule foregrounds conditional structure; a graph foregrounds nodes and relations; a statistical model foregrounds learned regularities; a Document foregrounds narrative or argument; a workflow foregrounds ordered action. Each can obscure what another exposes.

7.1 Formal semantics are local

A formal system may show that a conclusion follows from premises under declared semantics. It does not establish that the premises describe reality, that the vocabulary is adequate, that the transformation preserved all material meaning, or that anyone is authorized to rely or act.

7.2 Multiple Representations and interoperability

Propositions, procedures, examples, demonstrations, cases, simulations, measurements, narratives, models, and executable artifacts can express complementary aspects. SymK requires governed relations among them; it selects no universal ontology, logic, schema, inference method, lowering, or canonical package here.

Interoperability is achieved and maintained, not possessed by syntax alone. Shared identifiers can hide different classifications, institutional practices, time horizons, and consequences.

7.3 Human, Scientific, and Engineering Views

The current universal minimum comprises three Views:

- the **Human View**, concerned with meaning, experience, comprehensibility, participation, Power, burden, and consequence;
- the **Scientific View**, concerned with claims, Grounds, methods, Evidence, uncertainty, reproducibility, explanation, and challenge; and
- the **Engineering View**, concerned with requirements, models, interfaces, controls, operation, failure, maintenance, and evolution.

No View is sovereign. Human readability, formal inspectability, and machine inspectability can cross Views. Disagreement among Views must be represented and governed, not eliminated by silent translation. Every material Projection should declare source, target, purpose, additions, omissions, loss, reversibility, and validation route.

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8. Reasoning, assessment, reliance, decision, and application

Reasoning transforms or relates Representations through rules, algorithms, learned patterns, judgment, or mixed procedures. Review should preserve premises, source versions, methods, assumptions, uncertainty, alternatives, tools, models, participants, and the kind of output produced. Deduction, prediction, hypothesis, Answer, explanation, recommendation, decision support, decision, and action are not interchangeable.

Assessment is plural because different objects and questions require different criteria. Relevant dimensions may include:

- syntax or schema conformance;
- logical consistency or satisfiability;
- empirical comparison;
- benchmark and case performance;
- expert or peer review;
- usability and task fit;
- disagreement analysis;
- robustness and transfer;
- operational monitoring;
- security, privacy, legal, professional, or ethical review; and
- consequence assessment.

No single validator settles all dimensions. Formal consistency can preserve false premises. Consensus can be mistaken. Empirical performance can exploit a shortcut. User satisfaction can reward confident error. A pass on one dimension does not authorize reliance on another.

Reliance Authorization is distinct from epistemic status and assessment. A competent Authority may permit bounded reliance despite uncertainty, or prohibit use despite an apparently strong claim. A recommendation can inform a decision without becoming the decision; a decision can authorize action without guaranteeing execution or outcome.

Application changes the circumstances from which future observations are drawn. Consequences can motivate inquiry, challenge, correction, repair, retirement, or new Questions. They do not become Evidence or system Learning merely because they were recorded.

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9. A branching and recursive epistemic lifecycle

The lifecycle is a family of interacting phases, not an eleven-step universal workflow. Any phase can recur, branch, be skipped, fail, or reopen another phase:

1. purpose, Context, Scope, Domain, Value horizon, and Question framing;
2. participant, role, Bearer, affected-subject, Authority, and Responsibility identification;
3. inquiry, observation, elicitation, measurement, acquisition, or candidate generation;
4. contribution, claim, Ground, possible-Evidence, and Provenance construction;
5. Representation, Projection, comparison, translation, and loss declaration;
6. Reasoning, retrieval, computation, generation, judgment, and explanation;
7. assessment, disagreement, challenge, and uncertainty articulation;
8. Answer, recommendation, status, Reliance Authorization, decision, or other governed result;
9. application, action, omission, operation, and dependency;
10. observation of outcomes, consequences, harms, benefits, displacement, and residual effects; and
11. correction, repair, retraction, supersession, retirement, renewed inquiry, and possible Formation or Learning assessment.

These names provide review vocabulary. They do not require one order, artifact set, role topology, system architecture, or package. Each transition is typed, scoped, temporal, challengeable, and potentially lossy. No transition creates truth or Knowledge.

Possible Learning by a Human, artificial, collective, institutional, distributed, hybrid, or configured system must satisfy the accepted Learner and comparatively durable integrated-change conditions at the lowest adequate Bearer. Replacement, selection, redesign, current Context, retrieval, supervision, and environmental change remain alternatives to a system-Learning attribution. Exposure to feedback does not by itself establish Learning, Training, Education, memory, or capability continuity.

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10. Adjacent disciplines and jurisdiction

Knowledge Engineering interacts with adjacent disciplines without absorbing their jurisdiction.

10.1 Epistemology

Epistemology investigates Knowledge, truth, warrant, testimony, disagreement, and related questions. Knowledge Engineering constructs conditions and artifacts for epistemic work. It depends on epistemic distinctions without converting philosophical inquiry into an implementation schema.

10.2 Data and information engineering

Data engineering governs acquisition, storage, transformation, quality, movement, and availability of represented observations or records. Information architecture governs organization and retrieval for users and tasks. Neither manages “facts” while Knowledge Engineering alone manages “meaning.” Their objects and criteria overlap but do not collapse.

10.3 Software and AI engineering

Software engineering governs executable systems and their lifecycle. AI engineering governs model, data, evaluation, deployment, and operational arrangements for AI capabilities. Knowledge Engineering focuses on epistemic conditions across technical and nontechnical arrangements. A system can be well engineered as software and epistemically defective, or epistemically promising and operationally unsafe.

10.4 Knowledge management

Knowledge management concerns organizational practices for creating, sharing, retaining, and using Knowledge-related resources. Organizational-knowledge literature offers relevant models [Nonaka 1994], but organizational attribution, contribution, record, and individual Bearer must remain distinct.

10.5 Formation, Learning, Training, and Education

Formation is Foundational. Learning, Training, and Education are distinct supporting concepts. Epistemic conditions may contribute to any of them, but Knowledge Engineering neither universally depends on Education nor automatically educates or trains participants. Practice can be broader than Training, and capability continuity may be supported by documentation, redundancy, redesign, tools, recruitment, supervision, external service, or other routes.

10.6 Governance, law, ethics, and professions

Representation, access, classification, visibility, and automation can exercise or shape Power and Authority. SymK can require those effects to be represented and challenged. It does not thereby allocate legal rights, duties, remedies, professional competence, credentials, policy, or external jurisdiction.

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11. Bearers, participation, Cooperation, and generative systems

Knowledge may be propositional, practical, or direct/acquaintance Knowledge under the accepted baseline. Any attribution must identify the **lowest adequate Bearer**: the smallest stable system at which the relevant achievement, capability, integration, manifestation, and persistence are evidenced.

A Human participant can contribute without being the sole Bearer. A component can store or transform a Representation without knowing. A team can coordinate contributions without automatically becoming a collective Bearer. An institution can preserve records while current participants lack the relevant capability. A configured Human–AI system may be the adequate Bearer in one case while its Human, model, workflow, or institution is the adequate Bearer in another.

Terms such as *hybrid*, *distributed*, and *collective* describe configurations or candidate levels; they do not settle attribution. Member properties do not transfer to wholes, and whole properties do not transfer to members.

Cooperation is an actual, scoped relational process in which two or more distinguishable Participants make materially relevant Contributions that are mutually oriented toward a bounded shared undertaking and sufficiently connected for the Contributions to form part of that undertaking rather than merely coincide or aggregate. It is a thin relational occurrence and is not automatically beneficial, symmetric, voluntary, successful, Knowledge-producing, or the final objective. Contributions may support Knowledge at an evidenced Bearer without “creating new Knowledge” as an automatic collective effect.

Generative systems can contribute Questions, candidates, transformations, comparisons, retrievals, and critiques. Their outputs require the same claim, Ground, Evidence, Provenance, Representation, assessment, Authority, and consequence distinctions. Human approval does not erase model contribution or guarantee correctness. Model contribution does not erase Human or institutional Responsibility.

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12. Twelve explanatory obligations

The v0.1 `KE-01`–`KE-12` proposals are retained here as explanatory obligations. They are not candidate axioms, constitutional commitments, Ratified rules, or a universal system specification.

1. **Object boundary:** explain what epistemic conditions are engineered and what is not.
2. **Knowledge non-production:** prevent artifacts, labels, approvals, and computation from manufacturing Knowledge by declaration.
3. **Construction:** treat acquisition as situated construction with visible contribution and transformation.
4. **Representation:** declare the purpose, selection, omission, loss, and limits of each material Projection.
5. **Plural modes:** preserve propositional, practical, and direct/acquaintance Knowledge without forcing one form.
6. **Ground/Evidence/Provenance:** keep support, role attribution, lineage, assessment, and status distinct.
7. **Reasoning boundary:** preserve premises, method, uncertainty, alternatives, output kind, and scope.
8. **Validation plurality:** match assessment dimensions to objects and prevent one check from settling all others.
9. **Bearer and contribution:** identify the lowest adequate Bearer and prohibit member/component/system transfer.
10. **Authority and Responsibility:** type roles, preserve jurisdiction, and route challenge and change through competent procedure.
11. **Recursive evolution:** support correction, disagreement, supersession, retirement, consequence return, and possible Learning without assuming them.
12. **Formation boundary:** represent formative conditions and actual effects separately; impose no universal Education, Training, virtue set, or architecture-effect inference.

These obligations explain what a credible SymK account should make visible. Later standards may operationalize subsets under explicit Authority, Domain, Context, Scope, and change control.

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13. Counterexamples the account must survive

A Knowledge Engineering account is inadequate if it cannot distinguish at least these cases:

1. a true claim stored with no identified Bearer;
2. a false but internally consistent ontology;
3. a correct deduction from false premises;
4. a useful prediction without a warranted causal explanation;
5. an authentic source that is mistaken;
6. a generated answer that is correct by chance;
7. a generated answer with strong Grounds but prohibited use;
8. expert consensus contradicted by later observation;
9. a Document that preserves instructions while embodied competence has decayed;
10. a team whose members contribute but whose collective Knowledge is not evidenced;
11. a configured system that performs through retrieval or supervision without system Learning;
12. an assessment passed in the Engineering View but failed in the Human View;
13. a formally reversible Projection with practically irrecoverable omitted context;
14. an affected subject able to challenge but lacking veto or decision Authority;
15. adverse Evidence that requires urgent consideration but creates no new emergency power;
16. a consequence observation that is disputed and not yet Evidence;
17. a technically successful deployment that produces harmful Value consequences; and
18. a beneficial outcome that does not validate every premise or process that preceded it.

No single score or gate result resolves these examples.

14. Development-facing review vocabulary

The following questions form a non-container review envelope. They are not a schema, certification, canonical workflow, or claim that completeness guarantees Knowledge.

Accessible table projection: Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **Review concern:** Purpose and Value
  - **Questions to preserve:** What is being attempted, for whom, why, and under which competing Values and consequences?
- **Review concern:** Context, Scope, Domain
  - **Questions to preserve:** Where, when, for whom, and under which conditions does each claim or use apply?
- **Review concern:** Participants and affected subjects
  - **Questions to preserve:** Who contributes, interprets, assesses, decides, acts, bears burdens, benefits, or may challenge?
- **Review concern:** Bearer
  - **Questions to preserve:** Which lowest adequate Human, artificial, collective, institutional, distributed, hybrid, or configured system may know or learn?
- **Review concern:** Questions and claims
  - **Questions to preserve:** What is asked, asserted, denied, hypothesized, prescribed, recommended, or answered?
- **Review concern:** Grounds and Evidence
  - **Questions to preserve:** What can support or defeat which object; who attributes the Evidence role under which criteria?
- **Review concern:** Provenance
  - **Questions to preserve:** What are the origin, contribution, custody, transformations, versions, and omissions?
- **Review concern:** Representation and Projection
  - **Questions to preserve:** What is selected, added, omitted, strengthened, weakened, lost, reversible, or hard to recover?
- **Review concern:** Reasoning and tools
  - **Questions to preserve:** Which premises, methods, models, algorithms, judgments, and alternatives produce the result?
- **Review concern:** Assessment
  - **Questions to preserve:** Which Human, Scientific, and Engineering dimensions apply, and where do they disagree?
- **Review concern:** Authority and Responsibility
  - **Questions to preserve:** Who may propose, assess, authorize reliance, decide, act, change, supersede, or retire; who remains answerable?
- **Review concern:** Standing and challenge
  - **Questions to preserve:** Which standing/challenge locus applies, what response is owed by procedure, and what does not follow automatically?
- **Review concern:** Application
  - **Questions to preserve:** How can a Representation or result inform interpretation, recommendation, decision, action, or omission?
- **Review concern:** Consequences

    - **Questions to preserve:** What benefits, harms, burdens, displacement, dependencies, residuals, and unintended effects occurred?
- **Review concern:** Evolution
    - **Questions to preserve:** How are uncertainty, disagreement, correction, repair, versioning, supersession, decay, and retirement handled?
- **Review concern:** Formation and Learning
    - **Questions to preserve:** Which conditions existed, what subject-level effect is claimed, and which alternative explanations remain?

Passing a later authorized review can show that declared questions and controls were addressed. It cannot certify that Knowledge exists, every Bearer is competent, every use is authorized, consequences are beneficial, or the system is universally responsible.

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15. Implementation-facing requirements

Implementations derived from this paper should be able, proportionately to Context, Scope, risk, and Authority, to:

1. represent objects, claims, Grounds, possible Evidence, Evidence-role attributions, Provenance, assessments, statuses, and authorizations separately;
2. identify participants, contributions, Bearers, affected subjects, Authorities, Responsibilities, and temporal boundaries;
3. preserve source versions, transformations, model/tool versions, assumptions, uncertainty, disagreement, and alternatives;
4. declare Representation and Projection purpose, additions, omissions, semantic or evidential loss, and reversibility;
5. distinguish Question, Reasoning, Answer, explanation, recommendation, decision support, decision, action, outcome, and consequence;
6. support plural assessment dimensions and disagreements across Human, Scientific, and Engineering Views;
7. constrain access, disclosure, privacy, security, authorization, and use without treating control as truth;
8. route affected-subject challenge without presuming its unresolved Authority classification or external remedy;
9. allow Evidence to challenge claims and decisions while requiring competent procedure for artifact and Authority-source change;
10. route urgent adverse Evidence to urgent consideration by competent operational or external Authority without creating a new emergency power;
11. preserve correction, retraction, versioning, supersession, retirement, and dependency impact;
12. record observations and consequences without automatically promoting them to Evidence, Knowledge, Value, or Learning;
13. test Bearer level and alternatives before attributing collective, artificial, hybrid, or system Learning;
14. preserve Formation effects, dependency, deskilling, displacement, and capability-continuity questions without imposing Education; and
15. maintain machine inspectability, formal inspectability where used, and Human comprehensibility without treating any as a sovereign View.

These are design obligations, not a prescribed data model. Different Domains may satisfy them through different artifacts, roles, procedures, and levels of formality.

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16. Accepted answers and live remainders

Several v0.1 questions now have accepted minimum answers while retaining downstream work:

> **Accessible table projection:** Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **v0.1 inquiry:** What is directly engineered?
  - **Accepted minimum:** Epistemic conditions, not truth or Knowledge directly.
  - **Live remainder:** Domain-specific object boundaries and implementation patterns.
- **v0.1 inquiry:** Who or what can know?
  - **Accepted minimum:** Attribute at the lowest adequate Bearer under accepted Knowledge modes.
  - **Live remainder:** Evidence standards for difficult collective, distributed, artificial, and hybrid cases.
- **v0.1 inquiry:** How should tacit Knowledge be handled?
  - **Accepted minimum:** Explicitization is selective transformation, not lossless extraction.
  - **Live remainder:** Domain practices for demonstration, apprenticeship, simulation, and capability preservation.
- **v0.1 inquiry:** What counts as Evidence?
  - **Accepted minimum:** A role relative to a specified object, criteria, assessor, Context, Scope, and time.
  - **Live remainder:** Domain criteria, thresholds, and competent assessors.
- **v0.1 inquiry:** Can one Representation suffice?
  - **Accepted minimum:** No universal form is selected; Projections declare loss.
  - **Live remainder:** Interoperability standards and task-specific adequacy.
- **v0.1 inquiry:** How are Authority and challenge related?
  - **Accepted minimum:** Typed Authority remains distinct; standing/challenge is an unresolved locus; change follows competent procedure.
  - **Live remainder:** External rights, duties, remedies, and detailed challenge protocols.
- **v0.1 inquiry:** How should consequences return?
  - **Accepted minimum:** Through observation, interpretation, inquiry, assessment, challenge, and governed revision routes.
  - **Live remainder:** Causal methods, monitoring thresholds, repair, and operational procedures.
- **v0.1 inquiry:** What should participants or systems learn?
  - **Accepted minimum:** No universal Education or Learning requirement follows; any attribution needs subject-level Evidence.
  - **Live remainder:** Capability-continuity choices and Domain Formation requirements.

This paper creates no new tension or deferred question. It preserves the accepted answers and the live remainders routed through existing lineage, especially DQ-005, DQ-017, DQ-018, DQ-022, and later implementation, Domain, external-authority, and Human-review work.

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17. Conclusion

Knowledge Engineering began around building AI systems with Domain expertise and matured through a recognition that elicitation, modeling, Representation, organization, validation, and maintenance are inseparable practical concerns. SymK retains that history while enforcing a sharper ontological and governance boundary.

Knowledge is an epistemic achievement attributable to an adequate Bearer, not the contents of a knowledge base. Claims, Grounds, Evidence, Provenance, Representations, computations, assessments, approvals, records, and consequences can contribute to Knowledge-related work without becoming interchangeable. Engineering acts materially on the conditions and transitions among them while remaining fallible, scoped, and challengeable.

The result is neither a linear production pipeline nor a universal platform schema. It is a branching, recursive discipline of epistemic-condition design under declared Context, Scope, Domain, Authority, Responsibility, Representation, Value, and consequence. Cooperation can organize Contributions without guaranteeing Knowledge. Formation can occur without Education. Generative systems can propose without warrant. Evidence can challenge without directly rewriting an authority source. Human, Scientific, and Engineering Views can disagree without one becoming sovereign.

> **SymK uses Knowledge Engineering to make epistemic conditions inspectable, governable, revisable, and accountable while refusing to confuse engineered artifacts or procedures with truth, Knowledge, justified reliance, responsible action, or beneficial consequence.**

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Document governance note

This v0.2 Markdown is a governed review draft created in 2.1.10 Stream E and integrated across the four-paper set in Stream F through `SYMK-2X-EV-166`. It reconstructs the complete v0.1 paper under Stage-Accepted DR-009–018 while preserving v0.1 as historical epistemic evidence.

The paper explains accepted concepts and records implementation-facing consequences. It creates no new concept disposition, deferred-question answer, tension, Authority allocation, constitutional selection, Ratification, canonical schema, certification, publication acceptance, or project commit. No v0.2 PDF exists at this checkpoint. Visual publication work, complete Human review, and any publication acceptance require later separately authorized streams.